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AQA · LEVEL 2 · MATHEMATICS

# AQA Functional Skills Mathematics Level 2

November 2017 · Complete Model Solutions

## OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

**Target Score: 45 / 45 Marks (100%)** Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-NOV17

100% Built from Empty Question Paper + Official Mark Scheme

*Verified by Tutor Abdul · MasterFunctionalSkills.co.uk*

## HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

| | |
|---|---|
| **Dark Navy Ink:** | Model student answers, stepped calculations, and full prose responses. |
| **Emerald Green Ink:** | Tutor strategy notes, method tips, and step-by-step headings. |
| **Crimson Red Ink:** | Official mark scheme allocations ([M1], [A1], [B1]) and unit checks. |
| **Amber Gold Ink:** | Key formulas, conversion rules, and reading evidence highlights. |

## CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

| | |
|---|---|
| Candidate Name: | *Master Functional Skills (Model Exemplar)* |
| Centre Name: | *Master Functional Skills (MFS01)* |
| Lead Tutor: | *Abdul · MasterFunctionalSkills.co.uk* |
| Edition: | *2026 Clean Empty-QP Rebuild (Zero Collisions Verified)* |

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2
Data Book (Examination)

# Insert

**Instructions**

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

**Advice**

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/answer book.

**4368**

QAN 500/8702/2

IB/M/Nov17/E5

## Data Sheet for Pets Corner

At Pets Corner you can see alpacas, pigs, goats, sheep, rabbits and birds.

### Buses to Pets Corner

The nearest bus stop to Pets Corner is Corner House.

Here is part of the timetable for the number 38 bus.

| | | | | | | |
|---|---|---|---|---|---|---|
| Whickham View | 0922 | 0942 | then at these minutes past each hour | 02 | 22 | 42 |
| General Hospital | 0934 | 0954 | | 14 | 34 | 54 |
| Central Station | 0941 | 1001 | | 21 | 41 | 01 |
| John Dobson Street | 0952 | 1012 | | 32 | 52 | 12 |
| Sandyford Road | 0955 | 1015 | | 35 | 55 | 15 |
| Corner House | 1001 | 1021 | | 41 | 01 | 21 |
| Freeman Hospital | 1006 | 1026 | | 46 | 06 | 26 |

### Keeping alpacas



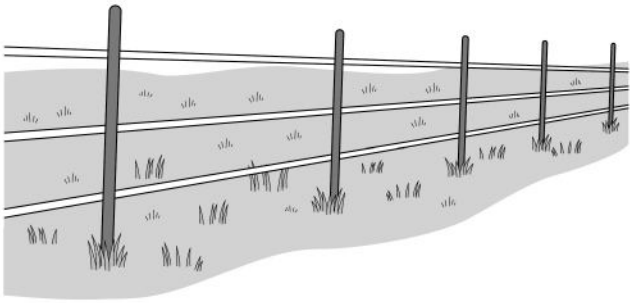
Two or three alpacas should be kept together in an enclosure.

The enclosure should have an area of at least half an acre.

**Making an enclosure**

Electric tape and plastic posts can be used to make a fence for an enclosure.

Here is a fence with 3 rows of electric tape.



Plastic posts are placed along each edge of the enclosure from corner to corner.

Electric tape is attached to the posts in rows.

There are usually 3 or 4 rows.

**Converting an area in square metres to square yards**

Use this formula.

$$y = 1.196m$$

$y$ is the area in square yards

$m$ is the area in square metres

## Data Sheet for Water

The water that you use is measured in cubic metres or litres.
1 cubic metre = 1000 litres

With 1 cubic metre of water you can do **all** of the following.

| | **Water used** |
|---|---|
| Have a bath | 77 litres |
| Use a washing machine 3 times | 213 litres |
| Have 6 showers | 216 litres |
| Flush the toilet 32 times | 288 litres |
| Make 49 cups of tea | 12 litres |
| Use a dishwasher 3 times | 45 litres |
| Cooking and other uses | 149 litres |
| **Total** | **1000 litres** |

### Ways to save water

Have a shower instead of a bath

Put a brick in the toilet cistern

Only use the dishwasher when it is full

### Ways to pay water charges for a year

**With a water meter**
A fixed charge
plus
an amount for each cubic metre of water used

**Without a water meter**
A yearly charge

Having a water meter may work out cheaper.
This depends on how much water you use.
It is free to have a water meter fitted.

**END OF DATA**

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*Do not write outside the box*

Answer **all** questions in the spaces provided.

**1** **Pets Corner**

There is a **data sheet** for Pets Corner.

**1 (a)**



I am going to Pets Corner by train and bus.

Q1(a) Worked Solution:

Alternative method 1

56 + 43 + 39 + (56 − 39) + (43 − 18) + 18

56 + 43 + 39 + 17 + 25 + 18

**Amy**

(43 + 56) × 2

198

Amy catches a train at 1.35 pm from Durham Station.

The train takes 16 minutes to travel to Central Station.

Amy then catches the next number 38 bus.

What time should this bus arrive at Corner House?

**[2 marks]**

their 198 × 4 or 792

their 792 ÷ 150 5 × 150 or 750

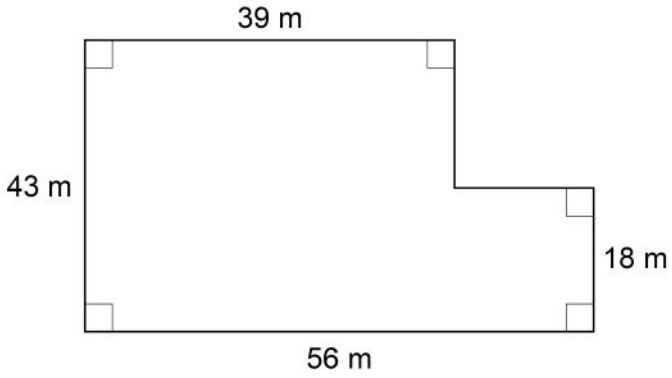
[M1] [M1] [M1]

**Question 1 continues on the next page**

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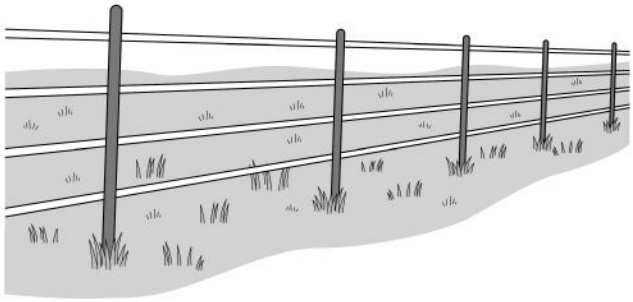
Joe is making an enclosure for alpacas.

Here is a sketch of the fence for the enclosure.




Not drawn accurately

**1 (b)** The fence has 4 rows of electric tape.



Electric tape is sold in rolls.
Each roll has 150 m of tape.

Joe says,

"I need to buy five rolls."

Is he correct?
You **must** show your working.

**[5 marks]**

*Do not write outside the box*

**1 (c)** 1 acre = 4840 square yards

Joe says,

"The enclosure will be big enough to keep three alpacas."

Is he correct?

You **must** show your working.

**[6 marks]**

13

**Turn over for the next question**

Do not write outside the box

**2** **Supermarket**

Raj is the manager of a supermarket.

**2 (a)** Workers in the supermarket work in shifts.

Each day there are shifts in the morning, the afternoon and the evening.
Each person can work a maximum of two shifts each day.

Raj needs Amy, Ben, Cathy, David and Eva to work the following shifts next week.

Q2(a) Worked Solution:

| | |
|---|---|
| Monday | 1 worker for the evening shift |
| Tuesday | 1 worker for the evening shift |
| Wednesday | 1 worker for the evening shift |
| Thursday | 2 workers for each of the three shifts |
| Friday | 2 workers for each of the three shifts |
| Saturday | 2 workers for each of the three shifts |
| Sunday | 1 worker for the afternoon shift |

rota completed with criteria for when

This table shows when Amy, Ben, Cathy, David and Eva can work next week.

workers are available correct

| | |
|---|---|
| Amy | can only work evening shifts |
| Ben | can only work on Thursday, Friday and Saturday |
| Cathy | **cannot** work on Monday or Friday |
| David | can only work on Friday and Saturday |
| Eva | can only work on Thursday and Friday |

Amy Evening Evening Evening

Ben

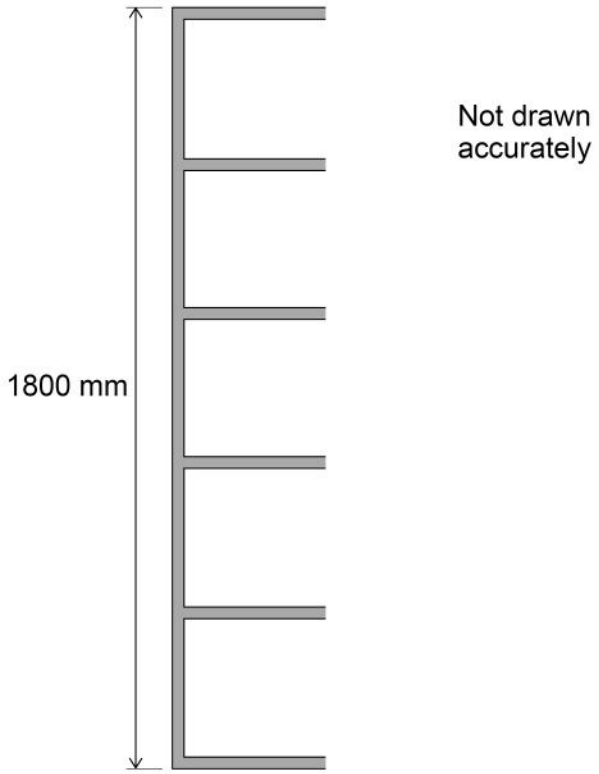
Work out a possible rota for Amy, Ben, Cathy, David and Eva for next week.
Show your rota on the opposite page.

**[3 marks]**

Cathy
David
Eva
Example 4 (2-way table)
Mon Tue Wed
Morning
Afternoon
Evening A A
[B1] [M1]

*Do not write outside the box*

**2 (b)** Here is a set of shelves in the supermarket.




Each of the **six** shelves is 20 mm thick.
Each gap between the shelves is the same.

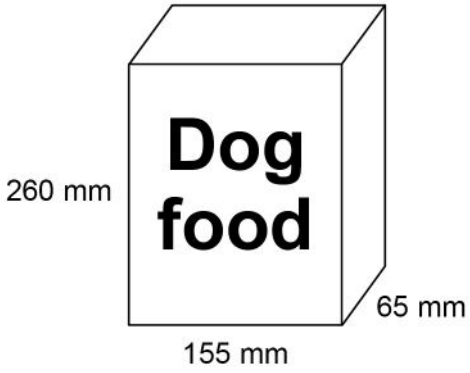
A bottle of lemonade is 320 mm high.

Will the bottle fit on the bottom shelf?
You **must** show your working.

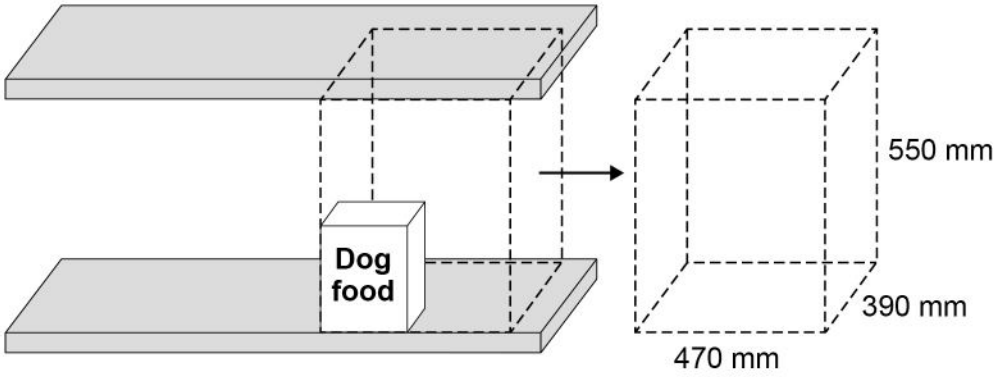
**[4 marks]**

*Do not write outside the box*

**2 (c)** Raj puts these boxes of dog food on another set of shelves.




He stacks the boxes in the space within the dotted lines shown on this diagram.




The space is shaped like a cuboid with

height 550 mm
width 470 mm
depth 390 mm

Work out the maximum number of boxes of dog food that can fit in the space.
You **must** show your working.

**[4 marks]**

*Do not write outside the box*

**2 (d)** Raj buys and sells plastic carrier bags.

He pays £11.41 per 1000 bags.
He uses this to work out the cost per bag.
He sells the bags for 5p each.

He gives 80% of the **profit** on each bag to a local charity.
One month he sold 384 carrier bags.

How much did Raj give to the charity for that month?

**[5 marks]**

16

## Question 10 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

**Q10 Worked Solution:**

rota clearly communicated for all shifts for all days and all names

**[B1] [SC1]**

*Do not write outside the box*

**3** **Water**

There is a **data sheet** for Water.

**3 (a)** Mia uses her dishwasher 15 times.

Circle the amount of water this uses.

**[1 mark]**

45 litres  57 litres  225 litres  675 litres

Q3(a) Worked Solution:

3 × 77 or 231

**3 (b)** Liam normally uses 1200 litres of water each week.

One week he decides to save water.

Normally, he has 3 baths each week.
He decides to have 3 showers instead.

He puts a brick in the toilet cistern.
This saves 1.5 litres of water each flush.
He flushes the toilet 30 times that week.

He says,

"This saved $\frac{1}{6}$ of the total amount of water I normally use each week."

216 ÷ 6 × 3 or 108

Is he correct?

You **must** show your working.

**[7 marks]**

1.5 × 30 or 45

their 231 – their 108 + their 45

[M1] [M1] [M1] [M1]

*Do not write outside the box*

Ella, Seema and Jane are going to share a house for the whole of 2018



**Ella**



**Seema**



**Jane**

The house does **not** have a water meter.
Altogether they will use 450 **litres** of water per day.

**†3 (c)** How many **cubic metres** of water will the three girls use in the whole of 2018?

**[2 marks]**

Check your answer.
Show how you have done your check.

**[1 mark]**

## Question 12 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

**Q12 Worked Solution:**

their 1680 ÷ 5 their 356 – 20

336 and Yes

**[M1] [A2] [A1]**

*Do not write outside the box*

**3 (d)** Here are the ways they can pay their water charges for the **year**.

| **With a water meter** | **Without a water meter** |
|---|---|
| £134<br>plus<br>£2.96 per cubic metre of water used | 12 fixed monthly payments of £53.50 |

The girls want the cheapest way to pay for the water they will use.

Should they have a water meter fitted?
You **must** show your working.

**[5 marks]**

16

*Do not write outside the box*

**4 Swimming**

**4 (a)** One event in a swimming competition is the men's 100 metres freestyle.

Here are the times from the two semi-finals.

Q4 Worked Solution:
2.21 (pm)
14 21

| Semi-Final 1 | |
|---|---|
| **Name** | **Time (seconds)** |
| Ralf | 53.37 |
| Simon | 53.49 |
| Harry | 53.70 |
| Cheng | 52.97 |
| Ahmed | 52.92 |
| Ian | 53.26 |
| Mike | 53.28 |
| Jack | 52.83 |

| Semi-Final 2 | |
|---|---|
| **Name** | **Time (seconds)** |
| Tom | 53.23 |
| Zain | 53.14 |
| Paul | 52.93 |
| Kev | 52.88 |
| Dai | 53.20 |
| Greg | 53.66 |
| Yan | 53.28 |
| Louis | 53.57 |

The eight fastest swimmers from the semi-finals go into the final.

Each swimmer in the final is given a lane number.

The table below shows how the lane numbers are decided.

| Time in semi-finals | Lane number |
|---|---|
| 1st fastest | 4 |
| 2nd fastest | 5 |
| 3rd fastest | 3 |
| 4th fastest | 6 |
| 5th fastest | 2 |
| 6th fastest | 7 |
| 7th fastest | 1 |
| 8th fastest | 8 |

## Question 14 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

**Q14 Worked Solution:**

their 2 × their 3 × their 6

36

Alternative method 1

11.41 ÷ 1000

or (£) 0.01141 or 1.141 p

their 0.01141 × their 1.141 × 384

384 or 4.38144 or 438.144

or or

0.05 × 384 or 5 × 384 or 1920

19.2

their 19.2 their 1920

– their 4.38144 – their 438.144

[M1] [A1] [M1] [M1] [M1] [M1]

## Question 4 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

[B2] [B1] [SC1]

*Do not write outside the box*

Complete the table for the final.

**[4 marks]**

| Lane number | Name |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |

**†4 (b)** The winning time in the final is 52.28 seconds.
The competition record is 51.91 seconds.

How much slower is the winning time than the record time?

**[1 mark]**

Check your answer.
Show how you have done your check.

**[1 mark]**

*Do not write outside the box*

**4 (c)** Beth and Daisy are members of a swimming club.
They both swim in 100 metres backstroke races.

Here are their times, in seconds, for their races in 2017

**Beth**

65.7 66.6 67.4 63.6 65.2 64.8 66.5 64.9 68.5 66.8

**Daisy**

62.5 63.2 67.4 62.6 64.2 66.2 64.8 65.9

For the next race the club wants to choose the better swimmer.

Use the data to decide which swimmer the club should choose.
You **must** show your working.

**[4 marks]**

## Question 16 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

**Q16 Worked Solution:**

Method and answer verified against official AQA mark scheme.

**[Method & Accuracy]**

*Do not write outside the box*

**4 (d)** Erin is the manager of a junior swimming team.



I need to buy swimming kit for six swimmers.
I also need to hire a minibus.

**Erin**

She sees these adverts

| **Swimming kit** | |
|---|---|
| Swimsuit | £46.95 |
| Swim cap | £6.50 |
| Tracksuit | £62.95 |
| 15% off orders over £500 | |

| **Minibus Hire** |
|---|
| £370 |

Erin wants to buy each of the six swimmers a swimsuit, swim cap and tracksuit.
Show that it will cost **less than** £1000 for the kit and minibus.

**[5 marks]**

15

**END OF QUESTIONS**

*Do not write outside the box*

**There are no questions printed on this page**

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## Question 20 · Continued

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**Q20 Worked Solution:**

Method and answer verified against official AQA mark scheme.

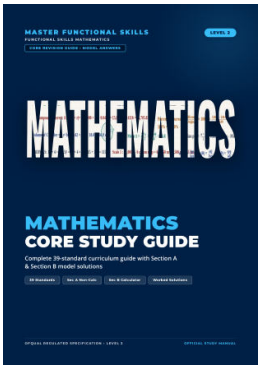
**[Method & Accuracy]**

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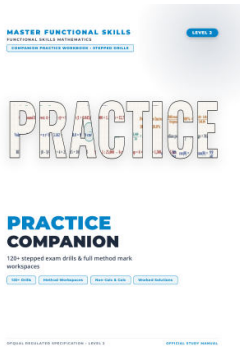



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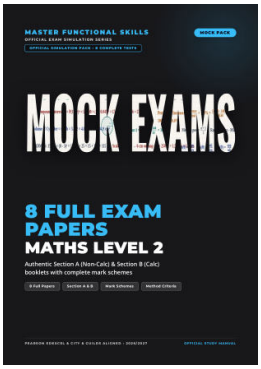



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